Strategy Land

Richard Rumelt's Web Journal on Good and Bad Strategy in Business, Politics, and Economics

How Piecemeal Change Can Make Things Worse

Incremental improvements in a chain-linked system reduces overall performance

Part 3 of Chains of Excellence [1 2 **3**]

January 6, 2011 By Richard Rumelt

One consequence of chain-link logic is *quality matching*. Quality matching means that the most economical approach is to balance the qualities of the chain-linked factors. That is, the presence of low quality factors reduces the incentive to invest in improving other factors.

The logic of quality-matching is not obvious at first glance, but it is by such deductions that scholars earn their keep. In this case, the intellectual work has been done by economists Michael Kremer and Sherwin Rosen.[1] The matching insight was hard-won because economics gained much of its analytical power by assuming vast pools of identical workers and capital equipment.

A century ago academic style required that a topic like quality matching be given an exhaustive treatment. The rise of Athens, feudal life, military organization, social structures in Indian life, manufacturing, and so on would each be investigated from the quality-matching perspective. Today, academic style asks that the problem be stripped to its bare essentials so that the central logic is exposed. We call this process model building because the purpose is not to describe the real world but, instead, to construct something so simple that its operation is transparent. The simplest situation in which quality matching occurs has a single product that is the joint outcome of two tasks. A model is just structure and logic, but it is a harmless pleasure to fix the structure within a special context. In this case, my context is Adam Smith's pin factory.

Adam Smith's magisterial book, *The Wealth of Nations*, wrought a revolution in thinking as profound as the simultaneous political revolution in the American colonies. Among Smith's many case examples, the most famous is that of the pin factory. One suspects that its fame rests at least partly on its position in his lengthy work—Volume I, Chapter 1, page 2.

Smith's pin factory was used to illustrate the gains to the division of labor. In his pin factory, "one man draws out the wire, another straights it, a third cuts it, a fourth points it, a fifth grinds it at the top for receiving the head; to make a head requires two or three distinct operations; to put it on, is a peculiar business, to whiten the pins is another; it is even a trade by itself to put them into the paper . . ."

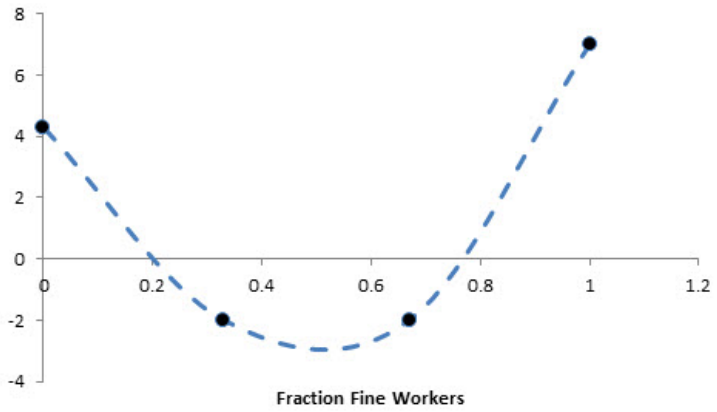
My model pin factory has only three tasks: (1) making *pointed wires*, (2) making *heads* with tiny sockets for the pointed wires, and (3) *joining* the pointed wires to the heads. Quality problems are only revealed in final inspection when some of the head fall off of the wires. This happens because the head was improperly formed, or because the wire was improperly finished, or because the joining operation took place improperly. It is impossible to fix such a pin and the components must be discarded. The only way to tell if either a head or a wire is defective is to try to join them into a pin.

In my model, pin workers come in two grades: Ordinary and Fine. Each type produces one batch of 100 heads, wires, or joins per production cycle. The Ordinary workers have 30% defect rates in their work. Fine workers have zero defects.

A small shop with a Fine worker in each of the three jobs makes 100 pins per cycle with no defects. Revenues are 1¢ per pin and wages are 31¢ per cycle for each fine worker (these are 18th century prices). Thus the shop's gross profit is 100 – 3 · 31 = 7¢ per cycle.

In a shop with all Ordinary workers, output is 34 pins per cycle. To see why, note that 70% of the heads are good, 70% of wires are good, and 70% of joins are good, so the chance that a good head, wire, and join come together is 0.7 · 0.7 · 0.7 = 0.34. Thus the three Ordinary workers are, together, about one-third as productive as the three Fine workers. Their wages are, therefore, about one-third as much, each receiving 10¢ per cycle.[2] Profits per cycle in the All Ordinary shop are 34.3 – 3 · 10 = 4¢ per cycle.

If the all-Fine shop makes 7¢ per cycle and the all-Ordinary shop makes 4¢ per cycle, then most people expect a shop with a mix of workers to have a profit rate somewhere in between 4¢ and 7¢. But this is not the case. What happens is surprising: adding a Fine worker to an all-Ordinary shop makes things worse. The basic result is shown below.



Pin Factory Profits (cents/cycle)

The shop makes a profit with NO Fine workers or with ALL Fine workers, but not with anything in between. With one Fine and two Ordinary workers, 49% of the pins are good, yielding a profit per cycle of 49 – 2 · 10 – 31 = -2¢. With one Ordinary and two Fine workers, 70% of the pins are good, yielding a profit per cycle of 70 – 10 – 2 · 31 = -2¢. The whole reason for the model is to shine an intense light on this effect.

Why does this happen? *With a mixed workforce the shop makes a loss because much of the value of a Fine wire-worker is wasted.* It is wasted because, although all of a Fine-worker's output is good, much of it must be discarded because of the Ordinary workers in the shop. More specifically, if there is one Ordinary worker, 30% of the other two Fine workers' output is discarded. If there are two Ordinary workers, 51% if the remaining Fine worker's output is discarded. *The Fine worker's contribution to output is limited by the qualities of the other workers*. Quantity is no substitute for quality because, no matter how many Ordinary workers are hired, much of the Fine wire-maker's output is still discarded.

Were this shop a complex company, operating as separate departments, the wire-making department would clearly have no incentive to change from Ordinary to Fine workers. Even though

high-quality companies are more profitable, *improving quality in just one department actually makes things worse.*

If two of the three "departments" are committed to using Fine workers, the remaining department then has a clear incentive to replace its Ordinary worker with a Fine worker, increasing profit to 10¢ per cycle. But with chain-link logic, this clear incentive appears only after *all of the other* departments are using improved processes. In this simple model, there are only two other departments. Were there twenty, none would have a clear incentive to improve unless most of the other nineteen had already done so.

In the model pin-factory, piecemeal change can make things worse rather than better. It does not matter whether the improvement in wire-making is mandated by the owner or whether the wire-making department autonomously tries to better itself. The result is the same—a fall in overall profit.

More generally, there are many reasons why it may be difficult to change things in large complex firms—why organizations get "stuck." Groups have entrenched interests that are not aligned with those of the company. Change threatens privilege, autonomy, and long-developed doctrine. Yet the lesson of the pin-factory is that even absent these social and political difficulties and frictions, piecemeal change may be unrewarding.

In the first entry in this series, Why Was GM Stuck?, the visiting GM executive described the problems as "Everything you have said. And More." Such dilemmas at companies like General Motors often have strong chain-link features. Increasing the quality of an automobile transmission does little good if the knobs fall off the dashboard and door panels continue to rattle. Improving fit and finish, along with the drive-train, may offer little improvement as long as the designers continue to produce pedestrian designs. Improving the look of the automobiles may only increase costs unless the complex technology of design-for-manufacturing is mastered. And so on.

Footnotes

1. Kremer, Michael, "The O-Ring Theory of Economic Development," *The Quarterly Journal of Economics*, **108**, 1993: 551-575. Rosen, Sherwin, "The Economics of Superstars," *American Economic Review*, LXXI, 1981, pp. 845-58. [↩]

2. If this were an academic journal, I would be asked to adjust these wages so that the profits of an all Fine and all Ordinary shops were the same, and then, perhaps, allow the entry of new shops until those profits were zero. Such additional complexity would not alter the conclusion but it would add complexity to the presentation. [↩]

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Tagged chain-link, change, management | 5 Comments

5 Responses to *How Piecemeal Change Can Make Things Worse*

Michael Lui [March 2, 2011 at 8:20 pm]
Great analysis, I am running into this exact issue at my client right now. We are trying to enhance the capabilities of one dept and are running into headwinds with other functions. Meanwhile, there is tension between showing meaningful benefits from transformational change and having the senior leadership support to truly orient the organization towards capturing sustained benefits. We are making headway, is there anyway to accelerate the process?

Reply

Richard Rumelt [March 3, 2011 at 2:27 am]
Thank you for the comment, Michael. Organizations get stuck because the overall payoffs to change don't appear until many changes have been made. In the pin factory story, the core of the dilemma is that one cannot measure the quality of the work of each employee, only the quality of the final product. The way through this is to focus on the change process itself and not the overall performance of the system. Senior management must measure and push for change in each unit and accept that the overall performance of the chain is not going to improve until most of the links have been fixed. This is a hard discipline, and it is why real change in larger organizations is so rare.

Reply

David Faulkner [August 9, 2011 at 12:00 am]

This is a good discussion which parallels Goldratt's Theory of Constraints. Improvement efforts targeting non-constraint processes or departments are not addressing the real problems in a given system and will drag cash generation down in the bargain.

Reply

Rohit [May 21, 2012 at 12:36 pm]

I notice that FINE workers enjoy a considerable premium in wages. Surely high-quality-lower-cost units would benefit the company?
Also, what wold happen if I were to reduce the gap in performance? I cant see it for myslef as I dont know what formula has been used for the graph

Reply

Richard Rumelt [May 21, 2012 at 2:44 pm]

It is a model designed to highlight this somewhat non-intuitive result.
3 fine workers produce 100 – 3*31 = 7 profit.
2 fine + 1 ordinary produce 0.7*100 – 2*31 – 10 = -2 profit.
1 fine + 2 ordinary produce 0.7*0.7*100 – 31 – 2*10 = -2 profit.
3 ordinary produce 0.7*0.7*0.7*100 – 3*10 = 4 profit.

The graph is just a smooth line drawn between these points.

Reply

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